6.2 Properties of Transition Elements (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	6. Inorganic Chemistry (A Level Only)
Topic	6.2 Properties of Transition Elements (A Level Only)
Difficulty	Medium

Time allowed: 70

Score: /58

Percentage: /100

Question 1a

Define a transition element.

[1 mark]

Question 1b

i)

NH_3 acts as a monodentate ligand. State what is meant by monodentate ligand.

[2]

ii)

Aqueous silver ions, Ag^+(aq), react with aqueous ammonia, NH_3(aq), to form a linear complex.

Suggest the formula of this complex, including its charge.

[1]

[3 marks]

Question 1c

There are two isomeric complex ions with the formula $[Cr(NH_3)_4Cl_2]^+$. One is green and the other is violet.

i)

Suggest the type of isomerism shown by these two complex ions.

[1]

ii)

Explain why these two complex ions are coloured and why they have different colours.

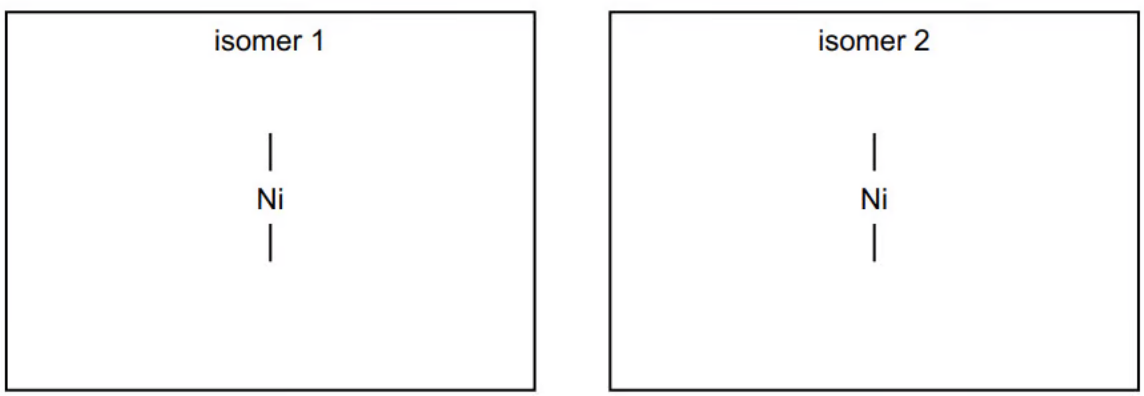
[4]

[5 marks]

Question 1d

The ligand ethane-1,2–diamine, $H_2NCH_2CH_2NH_2$, is represented by en.
Nickel forms the complex ion $[Ni(en)_3]^{2+}$ in which it is surrounded octahedrally by six nitrogen atoms.

Draw three-dimensional diagrams to show the stereoisomers of $[Ni(en)_3]^{2+}$.



[2 marks]

Question 1e

Ethane-1,2-diamine is a useful reagent in organic chemistry.

i)
Explain how the amino groups in ethane-1,2-diamine allow the molecule to act as a Brønsted-Lowry base.

[2]

ii)
Write an equation for the reaction of ethane-1,2-diamine with an excess of hydrochloric acid.

[1]

[3 marks]

Question 1f

i)
Under certain conditions ethane-1,2-diamine reacts with ethanedioic acid, HOOCCOOH, to form the polymer **Z**.
Draw the structure of polymer **Z**, showing two repeat units.

[2]

ii)
Name the type of reaction occurring during this polymerisation.

[1]

[3 marks]

Question 2a

Copper, vanadium and iron are all transition elements.

i)

Give the full electronic configuration of a Cu atom and a Cu^{2+} ion.

[2]

ii)

State four characteristic features of the chemistry of copper and its compounds.

[4]

[6 marks]

Question 2b

The chloride ion is a monodentate ligand. When concentrated hydrochloric acid is added to $[Cu(H_2O)_6]^{2+}$ (aq) ions the water ligands are replaced.

i)

Define the term *monodentate ligand*.

[1]

ii)

Write an equation for the reaction that occurs when concentrated hydrochloric acid is added to $[Cu(H_2O)_6]^{2+}$ (aq).

[1]

iii)

Draw the structure of the complex ion formed and name the shape it takes.

[2]

iv)

State the change in coordination number.

[1]

[5 marks]

Question 2c

Vanadium (V) oxide is the catalyst used in the contact process as shown by the reactions:

$$SO_2 + V_2O_5 \rightarrow SO_3 + V_2O_4$$

$$V_2O_4 + ½O_2 \rightarrow V_2O_5$$

i)

Using the equations, write an overall equation for the reaction.

[1]

ii)

Using the equations, explain why V_2O_5 is a catalyst.

[1]

iii)

Explain why V_2O_5 is able to act as a catalyst in this reaction.

[1]

[3 marks]

Question 2d

When iron(II) compounds dissolve in water they form $[Fe(H_2O)_6]^{2+}$, the hexaaquairon(II) complex.

i)

State the full electronic configuration of an iron(II) ion.

[1]

ii)

Predict the shape and bond angle of the hexaaquairon(II) complex.

[2]

[3 marks]

Question 3a

A solution is made by dissolving $CuSO_4{\bullet}5H_2O$ in an excess of aqueous ammonia. This solution contains the copper complex $[Cu(NH_3)_4]^{2+}$.

i)

Write an expression for the K_{stab} of $[Cu(NH_3)_4]^{2+}$.

K_{stab} =

[1]

ii)

State the colour of the solution of $[Cu(NH_3)_4]^{2+}$.

[1]

[2 marks]

Question 3b

The solution of $[Cu(NH_3)_4]^{2+}$ is heated gently in a fume cupboard so that NH_3 is released. Some NH_3 remains in solution and some forms NH_3 gas. The colour of the solution changes; a precipitate of $Cu(OH)_2$ forms and is collected.

A sample of $Cu(OH)_2$ is added to concentrated hydrochloric acid. A reaction takes place forming a coloured copper complex, **Y**.

A sample of $Cu(OH)_2$ is added to dilute sulfuric acid. A reaction takes place forming a coloured copper complex, **Z**.

$[Cu(NH_3)_4]^{2+}$, **Y** and **Z** are different colours.

Suggest an equation for the reaction of $[Cu(NH_3)_4]^{2+}$ to form $Cu(OH)_2$ as the aqueous solution of $[Cu(NH_3)_4]^{2+}$ is heated.

[1 mark]

Question 3c

Suggest an equation for the reaction of $Cu(OH)_2$ with concentrated hydrochloric acid, forming **Y**.

[2 marks]

Question 3d

Complete Table 3.1 with the colour and geometry of complex **Y** and the colour, geometry and formula of complex **Z**.

Table 3.1

	Y	**Z**
colour of complex		
geometry of complex		
formula of complex		

[2 marks]

Question 3e

Explain why complexes **Y** and **Z** are coloured and why their colours are different.

[5 marks]

Question 4a

This question is about the geometry and colours of transition metal complexes.

Explain why transition metals exhibit variable oxidation states compared to the elements in group 1.

[2 marks]

Question 4b

Transition metal compounds and ions are often coloured. For example, $[Cr(H_2O)_6]^{3+}$ is green.

Explain why $[Cr(H_2O)_6]^{3+}$ and other complex ions are coloured.

[3 marks]

Question 4c

Zinc and cobalt ions react with water to form the complexes $[Zn(H_2O)_4]^{2+}$ and $[Co(H_2O)_6]^{2+}$.

i)

Explain how water acts as a ligand in forming these complexes.

[2]

ii)

Predict the shape of $[Co(H_2O)_6]^{2+}$.

[1]

[3 marks]

Question 4d

Explain why solutions containing $[Co(H_2O)_6]^{2+}$ are coloured but solutions containing $[Zn(H_2O)_4]^{2+}$ are not.

[4 marks]

Model Answers: Medium

1a

a) A transition element is:

- (An element that can) form / has one or more stable ions with an incomplete d subshell; [1 mark]

[Total: 1 mark]

- This is a standard definition that you are required to know

1b

b)

i) The term monodentate ligand means:

- (A species that can) form one dative / coordinate bond; [1 mark]
- That uses a lone pair of electrons to form a dative / coordinate bond to a (metal) atom / ion; [1 mark]

ii) The formula of this complex, including its charge is:

- $[Ag(NH_3)_2]^+$; [1 mark]

[Total: 3 marks]

- Monodentate ligands can form only one dative bond to the central metal ion
- Examples of monodentate ligands are:
  - Water (H_2O) molecules
  - Ammonia (NH_3) molecules
  - Chloride (Cl^-) ions
  - Cyanide (CN^-) ions
- If asked to draw the complex don't forget to show the coordinate bond by using arrows which point to the metal ion as this is the direction of donation
- You should also know examples of bidentate and polydentate ligands

- The most common examples of a linear complex are a copper (I) ion, (Cu^+), or a silver (I) ion, (Ag^+), as the central metal ion with two coordinate bonds formed to two ammonia ligands
    - They have a bond angle of 180 ° and have two coordinate bonds
- NH_3 does not have a charge, so the overall charge of the silver ion, Ag^+, will remain

1c

c)

i) The type of isomerism shown by these two complex ions is:

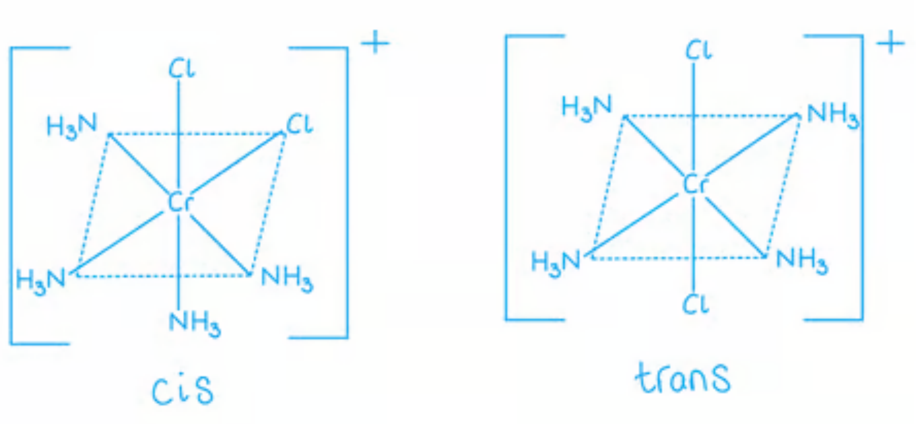
- Cis-trans
  OR
  Geometrical; [1 mark]

ii) These two complex ions are coloured and have different colours because:

- The d orbitals are split into 2 / different levels; [1 mark]
- (Colour is due to) absorption of light / photon (in the visible region); [1 mark]
- When an electron promotion / excited; [1 mark]
- The energy gap is different for the two complexes; [1 mark]

[Total: 5 marks]

- $[Cr(NH_3)_4Cl_2]^+$ can form two isomers
    - The dotted line represents the plane of the complex

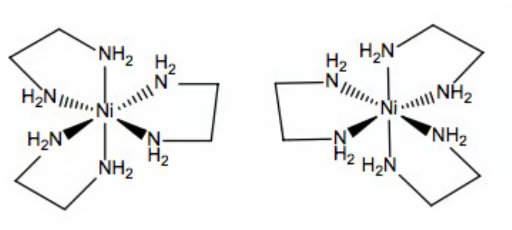


- Part (ii) is a very common question and so it is worthwhile learning this thoroughly

- the marking points will be the same!

1d

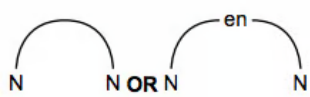
d) Three-dimensional diagrams to show the stereoisomers of $[Ni(en)_3]^{2+}$ are:



- First structure correct; [1 mark]
- Second correct
  AND
  Mirror image of the first structure; [1 mark]

[Total: 2 marks]

- Ethane-1,2-diamine, $H_2NCH_2CH_2NH_2$ is a bidentate ligand and it can be drawn relatively simply
  - Each end of the molecule has an amine group with two carbon atoms in the middle
  - Bidentate ligands can each form two dative bonds to the central metal ion
  - This is because each ligand contains two atoms with lone pairs of electrons
  - Each NH_2 group can donate a lone pair of electrons to the central metal ion
- To make two different isomers when you have only bidentate ligands, simply draw the mirror image of one
  - This is optical isomerism
- You are also allowed to use en as a bidentate ligand (instead of $H_2NCH_2CH_2NH_2$) as long is it correctly represented



1e

e)

i) The amino groups in ethane-1,2-diamine allow the molecule to act as a Brønsted-Lowry base because:

- The nitrogen lone pair; [1 mark]
- Accepts a proton / is a proton acceptor; [1 mark]

ii) An equation for the reaction of ethane-1,2-diamine with an excess of hydrochloric acid is:

- $H_2NCH_2CH_2NH_2 + 2HCl \rightarrow ClH_3NCH_2CH_2NH_3Cl$; [1 mark]

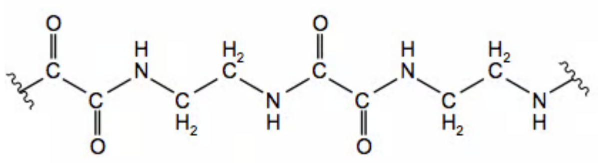
[Total: 3 marks]

- You need to know the definition of a Brønsted-Lowry acid and base and examples of each
  - Brønsted-Lowry acids donate a proton
  - Brønsted-Lowry bases accept a proton using its lone pair of electrons
- As ethane-1,2-diamine has two amine groups there are two places in which protons can be accepted
  - You can also use the ionic equation to score the mark
  - $H_2NCH_2CH_2NH_2 + 2H^+ \rightarrow (H_3NCH_2CH_2NH_3)^{2+}$

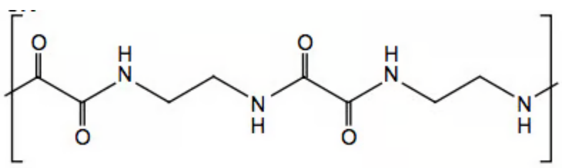
1f

f)

i) The structure of polymer **Z**, showing two repeat units



OR



- Amide bond / CONH; [1 mark]

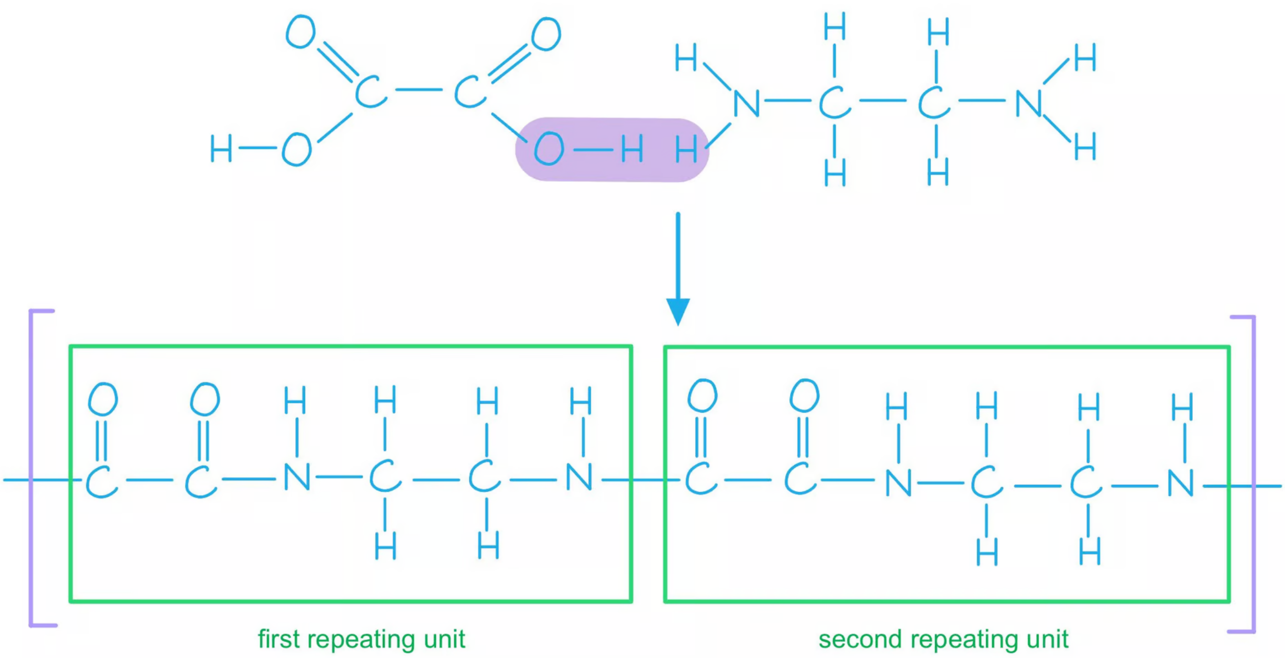
- Rest of the molecule with continuation bonds; [1 mark]

ii) The type of reaction occurring during this polymerisation is:

- Condensation; [1 mark]

[Total: 3 marks]

- Polyamides are also formed using **condensation** polymerisation
- We know the structure of ethane-1,2-diamine and ethanedioic acid so start by drawing these out
- You must show the correct amide link / bond and continuation bonds
- Double-check your polymer once you have drawn it as it is easy to miss out on the C=O, an H atom or even have an extra O atom in the polymer chain



2a

a)

i) The full electronic configuration of a Cu atom and a Cu^{2+} ion:

- $1s^22s^22p^63s^23p^63d^{10}4s^1$; [1 mark]
- $1s^22s^22p^63s^23p^63d^9$; [1 mark]

ii) Four characteristic features of the chemistry of copper and its compounds:

- Variable oxidation state; [1 mark]
- Form coloured ions/compounds; [1 mark]
- Form complexes / complex ions; [1 mark]
- Act as catalysts; [1 mark]

[Total: 6 marks]

- For the copper atom 3d and 4s can be switched around
- **Remember:** Copper has an unusual configuration because of the stability of filled and half-filled sub-shells
- The characteristic features are common to all transition metals

2b

b)

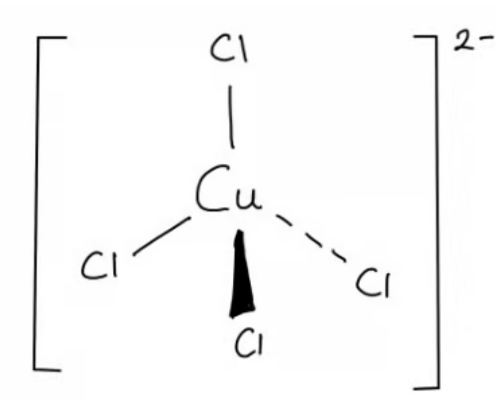
i) The term monodentate ligand means:

- A ligand that forms one coordinate bond / donates one lone pair of electrons to a metal atom or ion; [1 mark]

ii) The equation for the reaction when concentrated hydrochloric acid is added to $[Cu(H_2O)_6]^{2+}$ is:

- $[Cu(H_2O)_6]^{2+} + 4Cl^- \rightarrow [CuCl_4]^{2-} + 6H_2O$; [1 mark]

iii) The structure of the complex ion its shape:



- Correct structure

AND

Correct charge; [1 mark]

- Tetrahedral shape; [1 mark]

iv) The change in coordination number is:

- 6 to 4; [1 mark]

[Total: 5 marks]

- Ligands all have at least one lone pair of electrons to form a coordinate bond
  - The term 'mono' specifically indicates that one lone pair of electrons are involved
- In the ligand substitution reaction, the six water ligands are replaced by four chloride ions
  - The original shape of the complex was octahedral due to the six bonding pairs to the water ligands
  - The shape of the complex formed with chloride ligands is tetrahedral as it has four bonding pairs of electrons around the central metal ion
  - As the chloride ion is larger than water a change in coordination number is observed

2c

i) The overall equation for the reaction:

- $SO_2 + ½O_2 \rightarrow SO_3$
  OR
  $2SO_2 + O_2 \rightarrow 2SO_3$; [1 mark]

ii) V_2O_5 is a catalyst because:

- It takes part in the reaction but it is reformed / not used up; [1 mark]

iii) V_2O_5 is able to act as a catalyst in this reaction because:

- It has variable oxidation states; [1 mark]

[Total: 3 marks]

- To generate the overall reaction equation the two equations should be combined
  - $SO_2 + V_2O_5 + V_2O_4 + ½O_2 \rightarrow SO_3 + V_2O_4 + V_2O_5$
  - Anything present on both sides of the equation will cancel out - V_2O_5 and V_2O_4
- Catalysts speed up the rate of a reaction by providing an alternative pathway with lower activation energy - whilst they take part they are not used up, usually because they are regenerated
- Transition metals and their compounds have variable oxidation states which enable them to be catalysts

2d

i) The full electronic configuration of an iron(II) ion:

- $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$; [1 mark]

ii) The shape, and the bond angle of the hexaaquairon(II) complex:

- Octahedral; [1 mark]
- 90°; [1 mark]

[Total: 3 marks]

- The iron(II) ion will have lost two electrons from the atom configuration, losing the two electrons from the 4s subshell
- The complex is octahedral as there are 6 bonding pairs of electrons around the central atom which repel each other equally
- The bond angle within an octahedral molecule is 90°

3a

a)

i) The expression for the K_{stab} of $[Cu(NH_3)_4]^{2+}$ is:

- $K_{stab} = \dfrac{[Cu(NH_3)_4{}^{2+}]}{[Cu(H_2O)_6{}^{2+}][NH_3]^4}$; [1 mark]

ii) The colour of the solution of $[Cu(NH_3)_4]^{2+}$ is:

- <u>Deep / dark / royal</u> blue; [1 mark]

[Total: 2 marks]

- The $CuSO_4•5H_2O$ will be an aqueous solution, which means that you are working with the $[Cu(H_2O)_6]^{2+}$ complex ion
- The reaction changes from a blue solution of $[Cu(H_2O)_6]^{2+}$ to a deep blue solution of $[Cu(H_2O)_2(NH_3)_4]^{2+}$
  - $[Cu(H_2O)_6]^{2+} + 4NH_3 \rightarrow [Cu(H_2O)_2(NH_3)_4]^{2+} + 4H_2O$
- You can use the reaction equation to build the K_{stab} expression in a similar way to

writing a K_c expression

- **Remember:** Water / H_2O does not feature in the K_{stab} expression
- $K_{stab} = \dfrac{[Cu(NH_3)_4{}^{2+}]}{[Cu(H_2O)_6{}^{2+}]\,[NH_3]^4}$

3b

b) The equation for the reaction of $[Cu(NH_3)_4]^{2+}$ to form $Cu(OH)_2$ as the aqueous solution of $[Cu(NH_3)_4]^{2+}$ is heated is:

- $[Cu(NH_3)_4]^{2+} + 2H_2O \rightarrow Cu(OH)_2 + 2NH_4^+ + 2NH_3$
  OR
  $[Cu(NH_3)_4]^{2+} + 2H_2O \rightarrow Cu(OH)_2 + 2H^+ + 4NH_3$; [1 mark]

[Total: 1 mark]

- The first paragraph of the question gives you the information that you need for your answer:
  - *The solution of $[Cu(NH_3)_4]^{2+}$ is heated gently in a fume cupboard so that NH_3 is released.*
    - $[Cu(NH_3)_4]^{2+} + H_2O \rightarrow NH_3$
  - *Some NH_3 remains in solution and some forms NH_3 gas.*
    - $[Cu(NH_3)_4]^{2+} + H_2O \rightarrow NH_4^+ + NH_3$
  - *The colour of the solution changes; a precipitate of $Cu(OH)_2$ forms and is collected.*
    - $[Cu(NH_3)_4]^{2+} + H_2O \rightarrow Cu(OH)_2 + NH_4^+ + NH_3$
- The equation needs balancing:
  - The formation of $Cu(OH)_2$ will require $2H_2O$ for the two hydroxide ions
    - $[Cu(NH_3)_4]^{2+} + 2H_2O \rightarrow Cu(OH)_2 + NH_4^+ + NH_3$
  - To balance the charge on both sides requires $2NH_4^+$
    - $[Cu(NH_3)_4]^{2+} + 2H_2O \rightarrow Cu(OH)_2 + 2NH_4^+ + NH_3$
  - To balance the nitrogen atoms on both sides requires $2NH_3$
    - $[Cu(NH_3)_4]^{2+} + 2H_2O \rightarrow Cu(OH)_2 + 2NH_4^+ + 2NH_3$

3c

c) The equation for the reaction of $Cu(OH)_2$ with concentrated hydrochloric acid, forming

Y is:

$Cu(OH)_2 + 4HCl \rightarrow [CuCl_4]^{2-} + 2H_2O + 2H^+$

OR

$Cu(OH)_2 + 4Cl^- + 2H^+ \rightarrow [CuCl_4]^{2-} + 2H_2O$

- Correct $[CuCl_4]^{2-}$ complex including charge; [1 mark]
- The rest of the equation fully correct; [1 mark]

[Total: 2 marks]

- $Cu(OH)_2$ has a Cu^{2+} ion with two hydroxide / OH^- ions
- You need to know that the reaction of Cu^{2+} with HCl forms the $[CuCl_4]^{2-}$ complex
    - $Cu(OH)_2 + HCl \rightarrow [CuCl_4]^{2-}$
- This is an example of a neutralisation reaction so water will also be formed
    - $Cu(OH)_2 + HCl \rightarrow [CuCl_4]^{2-} + H_2O$
- There are 4 chlorine atoms on the right-hand side, so we need to use 4HCl
    - $Cu(OH)_2 + 4HCl \rightarrow [CuCl_4]^{2-} + H_2O$
- The two hydroxide ions react with two of the hydrogen ions from the HCl to form $2H_2O$
    - $Cu(OH)_2 + 4HCl \rightarrow [CuCl_4]^{2-} + 2H_2O$
- This leaves the remaining two hydrogen ions from the HCl unaccounted for
    - $Cu(OH)_2 + 4HCl \rightarrow [CuCl_4]^{2-} + 2H_2O + 2H^+$
- The addition of the $2H^+$ on the right-hand side also balances the charge

3d

d) The completed table with the colour and geometry of complex **Y** and the colour, geometry and formula of complex **Z** is:

	Y	Z
colour of complex	**yellow**	**(pale) blue**
geometry of complex	**tetrahedral**	**octahedral**
formula of complex		**$[Cu(H_2O)_6]^{2+}$**

- Any three correct answers; [1 mark]
- All correct answers; [2 marks]

[Total: 2 marks]

- Copper(II) transition metal chemistry has many shapes, colours, states and formulae that you are expected to know
  - You should know that $[CuCl_4]^{2-}$ is a yellow solution, one of the rare times that copper compounds are not blue
    - Due to the large size of the chloride ion, only four ligands can fit around the central copper(II) ion, which gives it a tetrahedral shape
- When $Cu(OH)_2$ is added to dilute sulfuric acid, it forms a solution of copper(II) sulfate
  - This means that there are sulfate / SO_4^{2-} ions and hexaaqua copper(II) ions / $[Cu(H_2O)_6]^{2+}$
  - These give the classic blue copper(II) colour and have an octahedral shape

3e

e) Complexes **Y** and **Z** are coloured and have different colours because:

- The d-orbitals split into two sets of energy levels of different energy; [1 mark]
- A wavelength / frequency / light / photon / hv is absorbed; [1 mark]
- Which causes electron(s) to be promoted / excited; [1 mark]
- The colour seen is complementary to the colour / energy absorbed; [1 mark]
- The d–d energy gap / ΔE is different for **Y** and **Z**
  AND
  Therefore, a different frequency / wavelength of light is absorbed for **Y** compared to **Z**; [1 mark]

[Total: 5 marks]

- The origin of colour in transition metal complexes is frequently identified by examiners as an area that many students struggle with
- Take the time to learn how the splitting of d-orbitals links to the frequency / wavelength of light being absorbed **AND** the frequency / wavelength / colour of light that is seen are linked to light energy and electrons

4a

a) Transition metals exhibit variable oxidation states in contrast to the elements in group 1 as:

Transition metals:

- (Contain) d and s orbitals which are close in energy
  OR
  (Successive) ionisation energies increase gradually; [1 mark]

In group 1 elements / alkali metals:

- The second electron is removed from a (much) lower energy level
  OR
  The removal of the second electron requires a significant increase in ionisation energy; [1 mark]

[Total: 2 marks]

- You should be able to describe the ability of transition metals to form variable oxidation states in terms of successive ionisation energies
- This is in contrast to the s-block metals, which have only one fixed oxidation state
  - For example, calcium is an alkaline earth metal and occurs with a +2 oxidation state in its ion and compounds
  - In contrast, the transition metal titanium occurs with oxidation states of +2, + 3, and + 4

4b

b) Complex ions such as $[Cr(H_2O)_6]^{3+}$ are coloured as:
Alternative 1:

- Ligands cause the splitting of 3d orbitals into two energy levels; [1 mark]
- Energy from visible light promotes an electron from a lower energy level to a higher energy level;
  OR
  Colour is due to an electron transition between (split) d orbitals; [1 mark]
- The energy absorbed is in the visible region of the spectrum
  AND
  (So) the complementary colour is expressed **OR** the colour is due to the remaining frequencies / wavelengths; [1 mark]

Alternative 2:

- Complexes have two sets of d orbital(s) of different energy
  OR
  d-orbitals split into two sets (of orbitals); [1 mark]
- Visible light is absorbed
  AND
  The complementary colour is observed; [1 mark]
- Electron(s) are promoted / excited to a higher (d–) orbital; [1 mark]

[Total: 3 marks]

- This is in contrast to the s-block metals, which have only one fixed oxidation state
  - For example, calcium is an alkaline earth metal and occurs with a +2 oxidation state in its ion and compounds
  - In contrast, the transition metal titanium occurs with oxidation states of +2, + 3, and + 4

4b

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Alternative 1:

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  Colour is due to an electron transition between (split) d orbitals; [1 mark]
- The energy absorbed is in the visible region of the spectrum
  AND
  (So) the complementary colour is expressed **OR** the colour is due to the remaining frequencies / wavelengths; [1 mark]

Alternative 2:

- Complexes have two sets of d orbital(s) of different energy
  OR
  d-orbitals split into two sets (of orbitals); [1 mark]
- Visible light is absorbed
  AND
  The complementary colour is observed; [1 mark]
- Electron(s) are promoted / excited to a higher (d–) orbital; [1 mark]

[Total: 3 marks]

- Questions about the origins of colour in transition metal complexes are typically for these same three marks
- So, it is worth taking the time to learn these points

4c

c)

i) Water can act as a ligand because:

- Water has a non-bonding / lone pair of electrons; [1 mark]
- (Which it) donates to the metal ion
  OR
  (Allows it to) form a co-ordinate / dative (covalent) bond with the metal ion; [1 mark]

ii) The shape of $[Co(H_2O)_6]^{2+}$ would be:

- Octahedral; [1 mark]

[Total: 3 marks]

- You should be able to describe what a ligand is and how it forms coordinate bonds within a complex ion
- A Lewis base is a species which can bond by donating a lone pair of electrons
  - We can describe ligands as Lewis bases
- Since $[Co(H_2O)_6]^{2+}$ has 6 water ligands, its shape has to be octahedral to spread the co-ordinate / dative covalent bonds as far apart as possible

4d

d) Solutions containing $[Co(H_2O)_6]^{2+}$ are coloured but solutions containing $[Zn(H_2O)_4]^{2+}$ are not due to:

- In transition metal complexes the d orbitals / levels are split (into two levels of different energy); [1 mark]
- In cobalt, the d levels are partially filled
  AND
  In zinc, they are completely filled; [1 mark]
- Visible light is absorbed when electrons move from lower to higher energy levels / transition between d orbitals (in cobalt); [1 mark]
- Transmitted light is the complementary colour; [1 mark]

[Total: 4 marks]

- You should know that zinc is not a transition metal
  - It has a complete d-sublevel, $[Ar]3d^{10}4s^2$

  - Its ion, Zn^{2+}, has the electron configuration, $[Ar]3d^{10}$
- Zinc will, therefore, not display the properties associated with transition metals such as forming coloured compounds
  - Compounds of zinc(II) are usually colourless
  - Unless the ligands in the complex have a chromophore (group of atoms responsible for the absorption of electromagnetic radiation), which can absorb in the visible region of the electromagnetic spectrum